

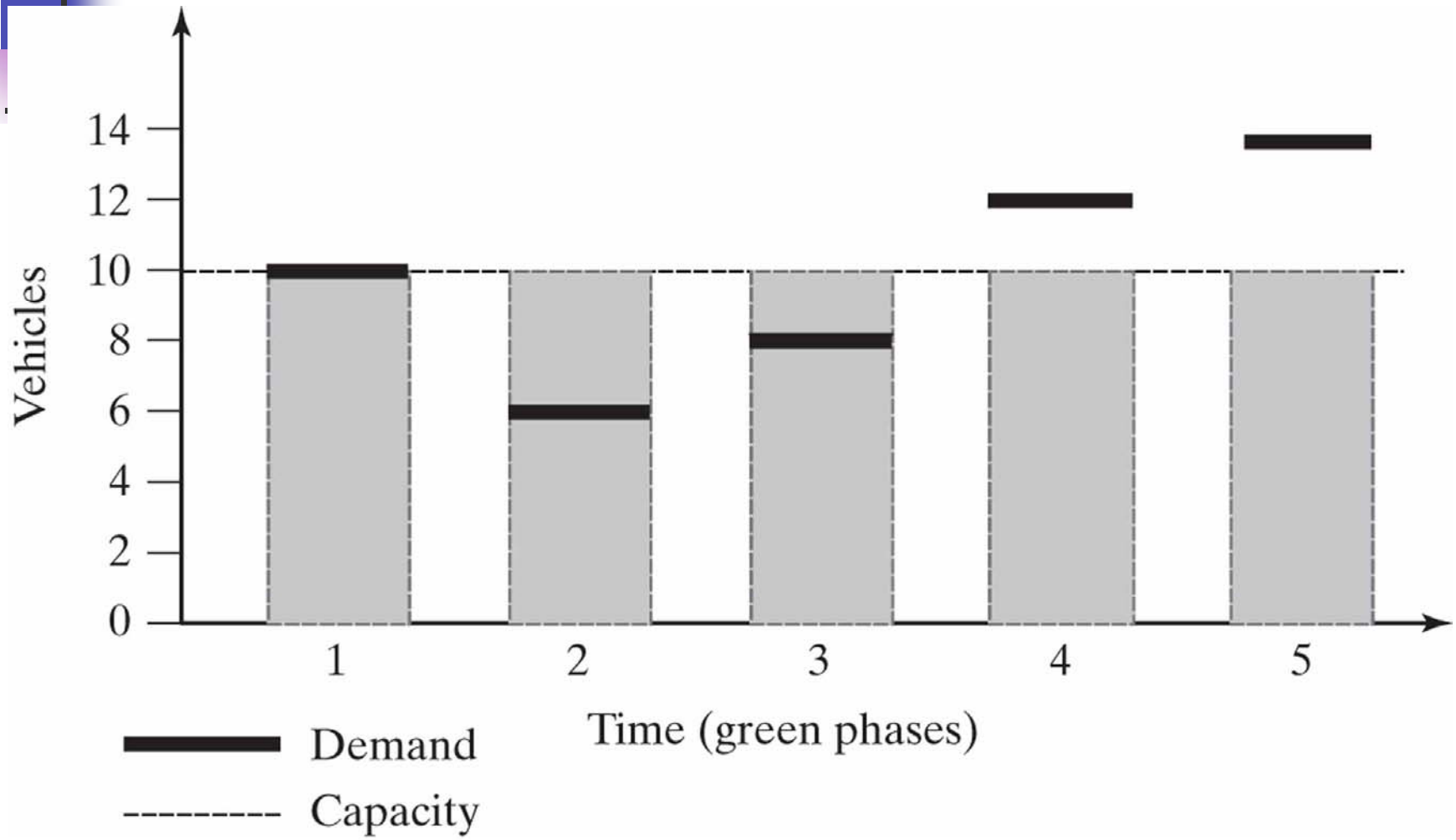
Chapters 22

Fundamentals of Actuated Signal Timing

Learning Objectives

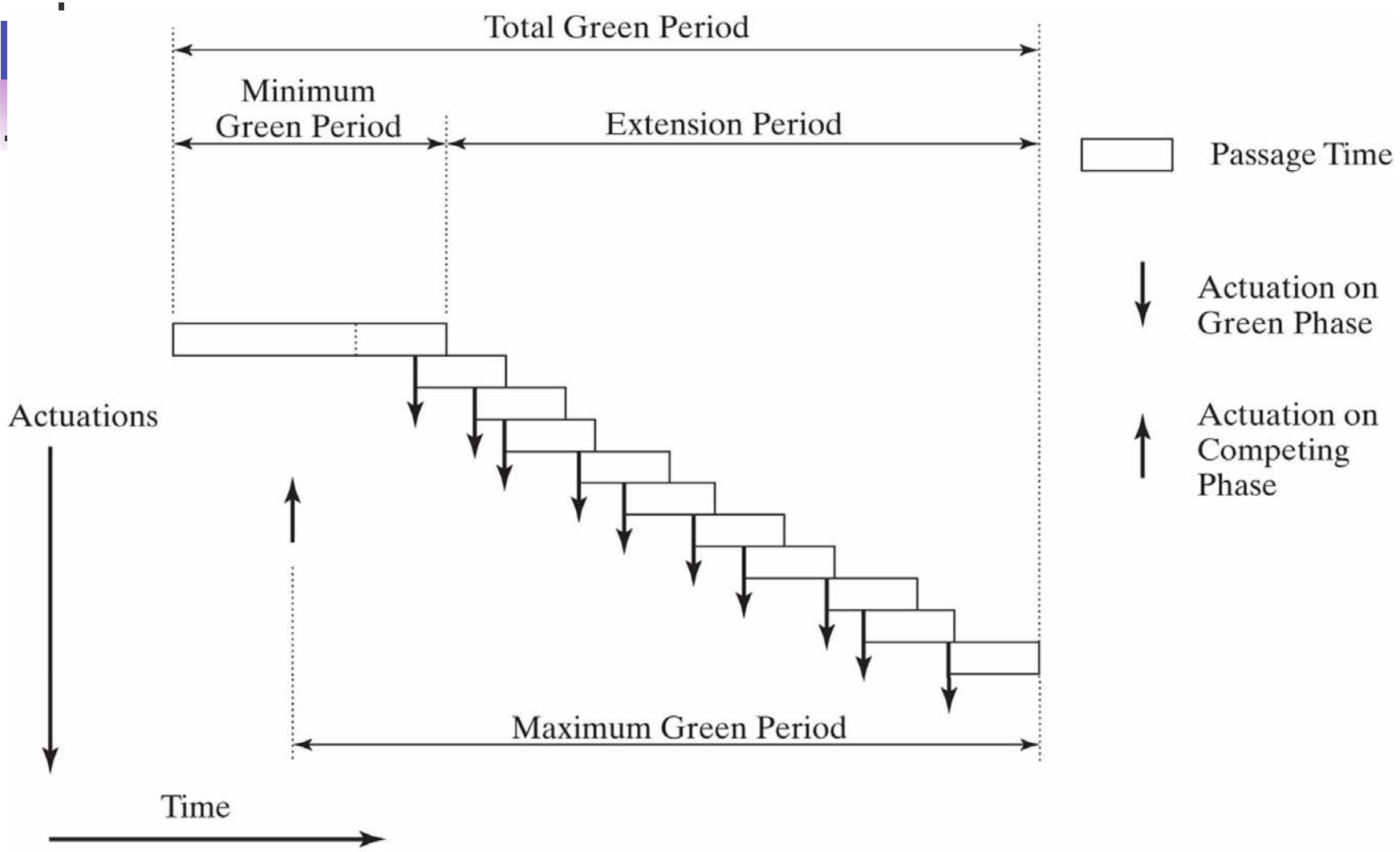
- Determine actuated signal control settings.

Introduction



Concept: Duration(s) of one or more green signal phases change in response to real time traffic demand.

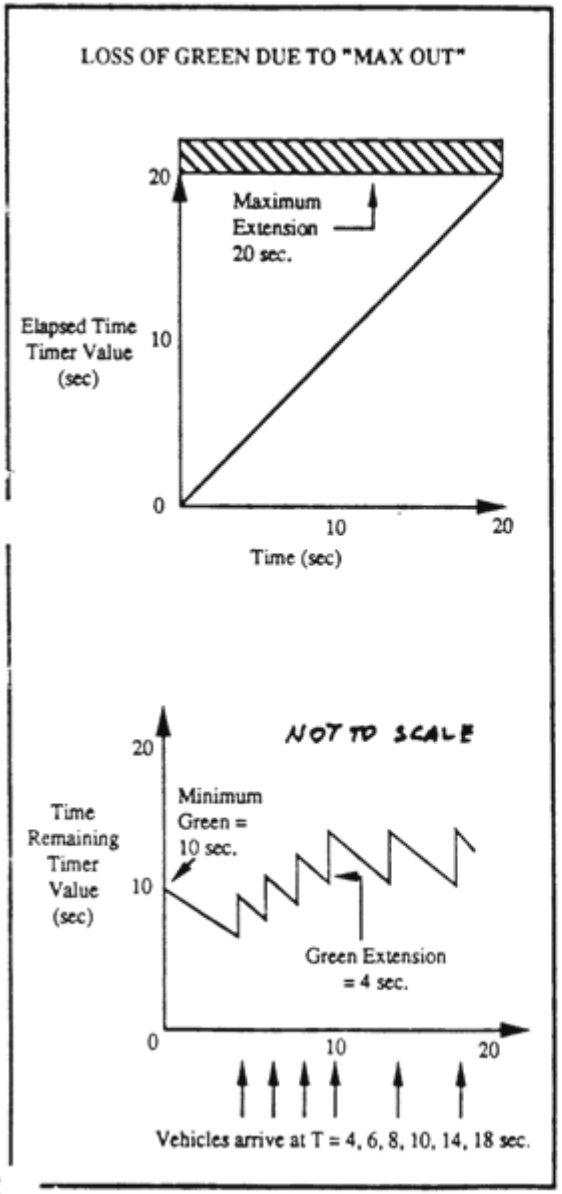
Figure 22.3 Operation of an Actuated Phase

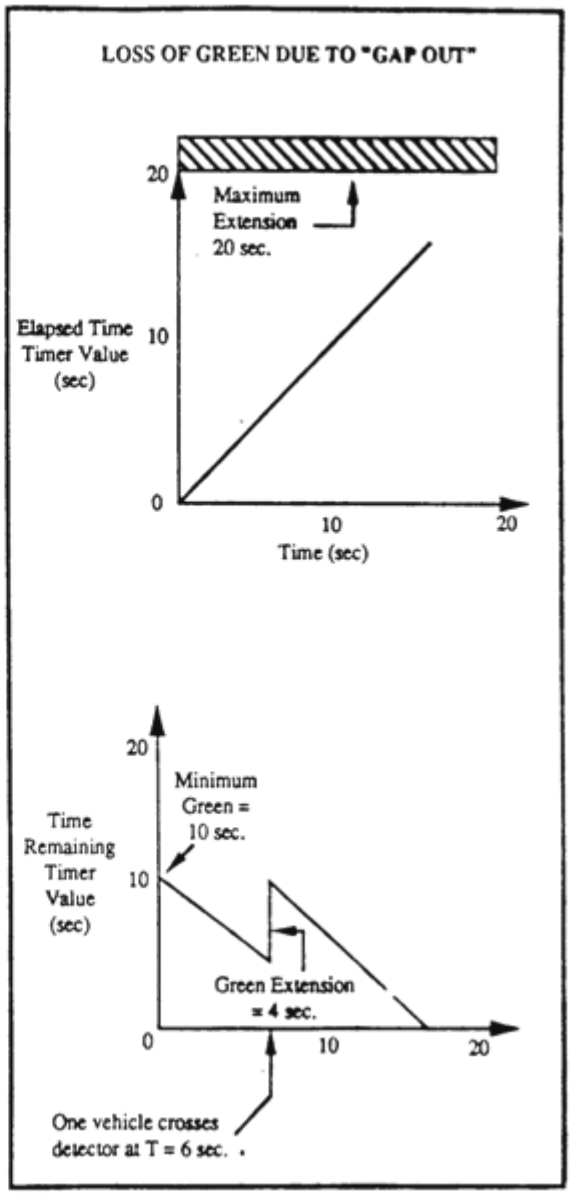


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Actuated Controller Operational Summary

- Every actuated phase has at least two timers that control the duration of the green phase.
  - (1) The elapsed time timer value starts at zero at the beginning of the phase and increases with time into the phase. If its value reaches the value set as the *maximum extension*, the green will be terminated. This loss of green is termed a "Max Out."
  - (2) The time remaining timer starts with a value of time as specified by the *minimum green* setting at the beginning of the phase. The value on the timer decreases with time into the phase unless a vehicle passes over the detector(s) attached to this phase in which case its value increases by the amount of the *green extension setting*. If no vehicle passes over the detector before its value reaches zero, the green phase is terminated. This loss of green is called a "Gap Out."


LOSS OF GREEN DUE TO "MAX OUT"
20
Maximum
Extension
20 sec.
Elapsed Time
Timer Value
(sec)
10
0
10
20
Time (sec)
NOT TO SCALE
20
Minimum
Green =
10 sec.
Time
Remaining
Timer
Value
(sec)
10
Green Extension
= 4 sec.
0
10
20
Vehicles arrive at T = 4, 6, 8, 10, 14, 18 sec.


LOSS OF GREEN DUE TO "GAP OUT"
20
Maximum
Extension
20 sec.
Elapsed Time
Timer Value
(sec)
10
0
10
20
Time (sec)
20
Minimum
Green =
10 sec.
Time
Remaining
Timer
Value
(sec)
10
Green Extension
= 4 sec.
0
10
20
One vehicle crosses
detector at T = 6 sec.

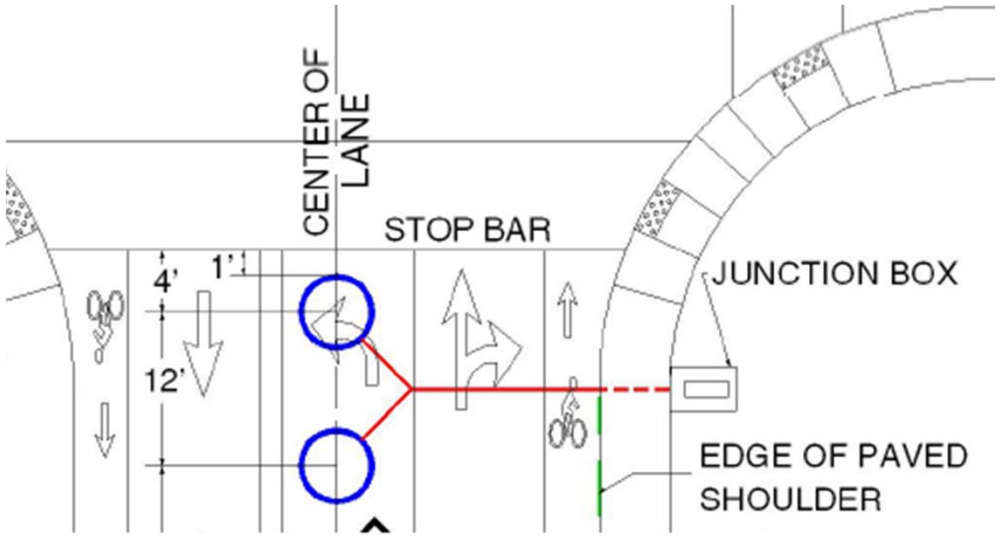
Fully vs. Semi-actuated

- Fully actuated: detectors provided on all approaches (phases)
  - Essentially equal volumes on both streets,
  - *Green dwells where last called*.
- Semi-actuated: detectors provided on less important intersection approaches only
  - One street clearly major,
  - *Green dwells on major or undetected street*

Detection techniques

- Inductance loop detectors, probably 85-90% of the market.
- Video detection, likely to grow in market share.
- Many others:
  - Audio
  - Electro-magnetic radiation
  - Electronic (passive and active systems)
  - Pressure sensitive

Example of loop detectors



(a) Schematic design of loop detector design



(b) Physical representation of loop detectors

Actuated control settings

- Settings for each actuated phase
  - 1) Maximum green time (same as pre-timed)
  - 2) Minimum green time
  - 3) Green extension or passage time
  - 4) Recall option
- Settings for all phases (actuated or not)
  - 1) Yellow or change interval (same as pre-timed)
  - 2) All-red or clearance interval (same as pre-timed)

Maximum green time and critical cycle

- A “critical cycle” is a cycle that has the highest cycle length.
- For a fully actuated signal, the critical cycle is one in which each phase reaches its maximum green time.
- For a semi-actuated signal, the critical cycle involves the maximum green time for the minor street and the minimum green time for the major street.
- The maximum green times for actuated phases and the minimum green time for the major street with semi-actuated control are found by determining the cycle length and green splits as done for a pre-timed signal.

Minimum Green Times

- The minimum green time is based on the type and location of detectors:
  - Point detectors

$$G_{min} = l_1 + 2.0 * \text{int}\left[\frac{d}{25}\right]$$

where

G_{min} = minimum green time, s

l_1 = start-up lost time, s

d = distance between detector and stop line, ft

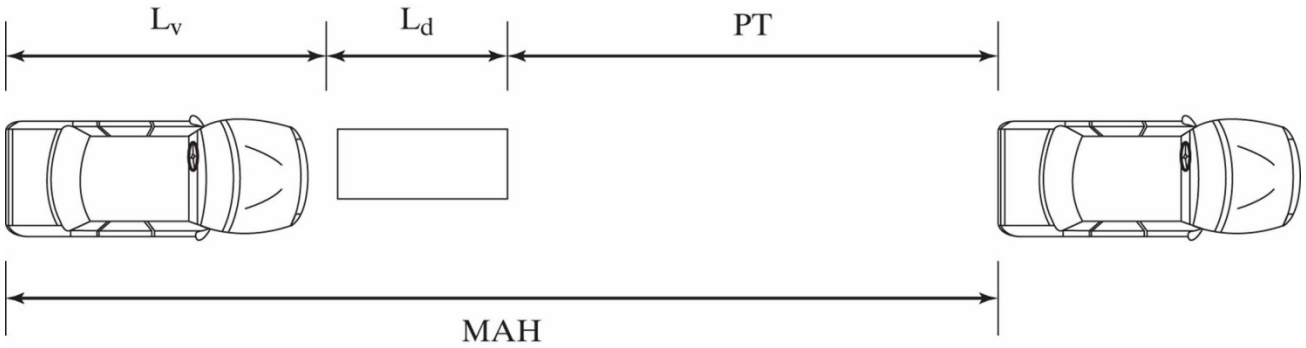
  - Presence detectors

$$G_{min} = l_1 + 2 \times n$$

where

n = number of vehicles stored in the detection area, vehs

Passage time (for presence detection)



- The key variable is the *maximum allowable headway* (MAH) or gap that will retain the green for a detector in a single lane.

$$PT = MAH - \frac{L_v + L_d}{1.47 S_a}$$

where

PT = passage time, s

S_a = average approach speed, mi/hr

L_v = length of vehicle, ft (default is 20 ft)

L_d = length of detection zone, ft

Passage time (for point detection)

- For point detection, L_d is 0 and L_v is irrelevant.
- Therefore, PT = MAH.

Minimum passage time

- Regardless of the type of detector being used, the passage time has a minimum value based on the location of the point detector or the front edge of a presence detector.
- The passage time must be at least large enough to allow a vehicle traveling at the 15th percentile approach speed to traverse the distance between the detector (or front edge of a presence detector) to the stop line:

$$PT_{\min} = \frac{d}{1.47S_{15}}$$

Source: Traffic Detector Handbook

Point detector location

Equation 22-1 can be used to determine the detector location given a desired Minimum green time. In practice, tables similar to the following are used.

$$G_{\min} = l_1 + 2.0 * \text{int}\left[\frac{d}{25}\right]$$

Approach speed		Detector setback from stopline to leading edge of inductive-loop detector		Minimum green time	Passage time
mi/h	km/h	feet	meters	seconds	seconds
15	24	40	12	9	3.0
20	32	60	18	11	3.0
25	40	80	24	12	3.0
30	48	100	30	13	3.5
35	56	135	41	14	3.5
40	64	170	52	16	3.5
45+	72+	Volume density or multiple inductive-loop detectors recommended.			

Some agencies limit the distance to 120 ft because of the long minimum green. This type of detection does not screen out false calls for green as occurs with right turns on red.

Presence detector location

- Typically, presence detectors are located very close to the stop line.
- This type of detection eliminates many of the false calls for green and thus avoids unnecessary green signals in approaches with no waiting vehicles.



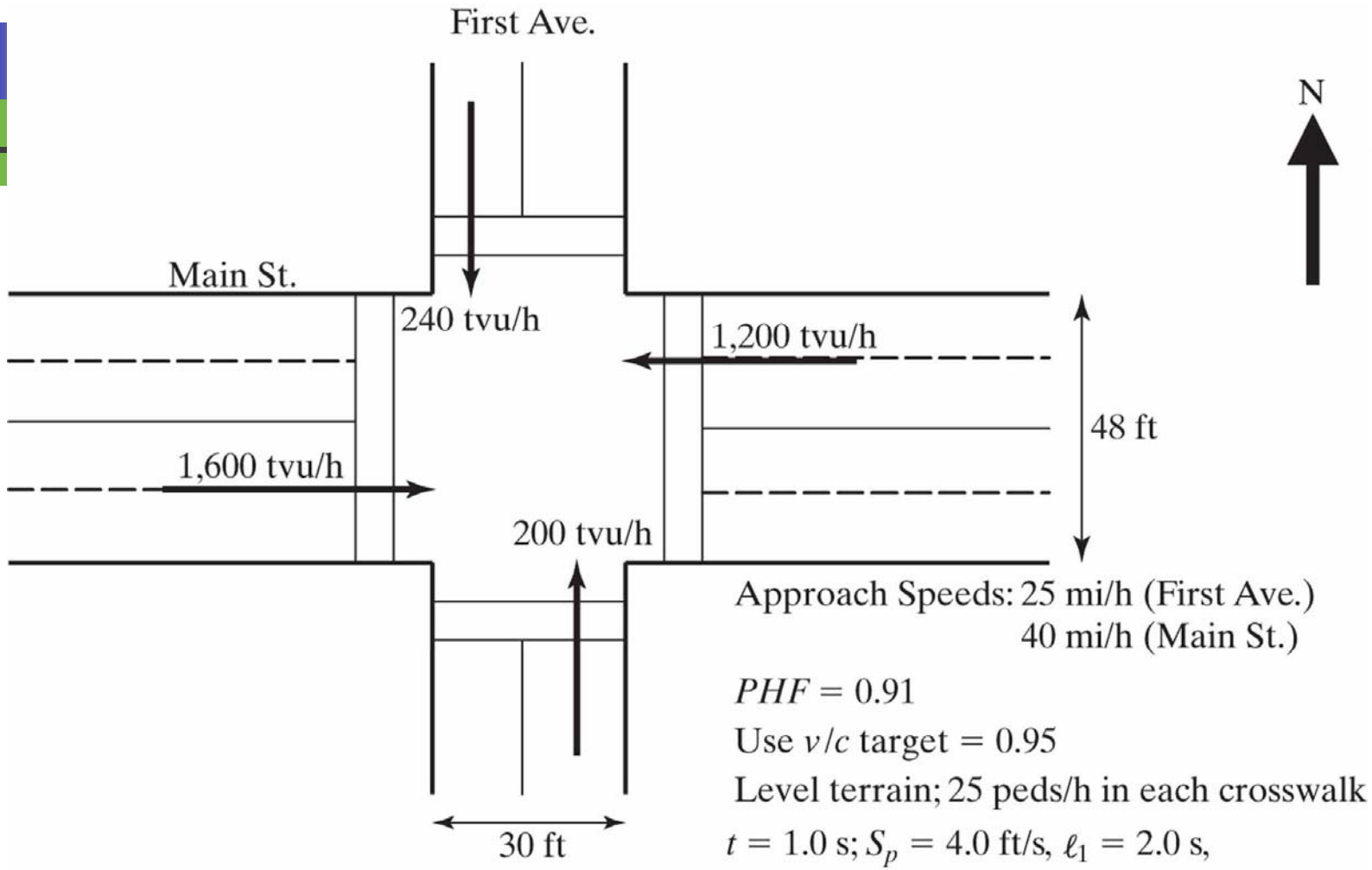
Recall features

- Recall settings determine what the signal controller will do when there is no demand present on one or more approach lane groups. A recall setting can be “off” or “on” for each phase. When “on,” there are several different options:
  - Minimum recall: when engaged, the minimum recall places a “call” on the designated phase, even if there is no demand present. If there is no demand anywhere, in the intersection, this would force the controller to initiate a green on the designated phase for a duration of at least the minimum green. Common practice is to engage the minimum recall on major street through movements.

Recall features

- Maximum recall: when engaged, it causes the controller to place a continuous call on the designated phase, even if there is no demand present. It forces every designated green phase to extend to its maximum green. This option is not often used but is appropriate if fixed-time operation is desired for some period of time, if vehicle detection is out of service, or if gap out is not desired.
- Pedestrian recall: when engaged, the controller places a continuous call for service on the designated phase. It forces the minimum pedestrian green to be implemented during every green phase. It is most often used where pedestrian actuation buttons are out of service or during time periods when large numbers of pedestrians are present.
- Soft recall: is almost the same as minimum recall, with one important difference: soft recall will discontinue its call for the green if there are conflicting calls at the intersection generated by actual vehicle demand. The end result is that phases set to soft recall can be skipped if there is no actual demand for the recall phase. With minimum recall, the recall phases are never skipped, they are always serviced even if there is no actual demand.

Example: A Semi-Actuated Signal



Approach Speeds: 25 mi/h (First Ave.)
40 mi/h (Main St.)

$PHF = 0.91$

Use v/c target = 0.95

Level terrain; 25 peds/h in each crosswalk

$t = 1.0$ s; $S_p = 4.0$ ft/s, $\ell_1 = 2.0$ s,

$e = 2.0$ s, $a = 10$ ft/s^2, $L = 20$ ft.

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